rashiga@localhost:~

```
[rashiga@localhost ~]$ vi prodcons.c
[rashiga@localhost ~]$ gcc prodcons. c -o prodcons
[rashiga@localhost ~]$ ./prodcons
Produced: 0
Consumed: 0
Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
Produced: 6
Consumed: 6
Produced: 7
Consumed: 7
Produced: 8
Consumed: 8
Produced: 9
Consumed: 9
```

rashiga@localhost:~ — vi prodcons.c

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>
#define BUFFER_SIZE 5
int buffer[BUFFER_SIZE];
int in = 0, out = 0;
sem_t mutex, empty, full;
void* producer(void* arg) {
    int item;
    for(int i = 0; i < 10; i++) {
        item = i;
        sem_wait(&empty);
        sem_wait(&mutex);
        buffer[in] = item;
        printf("Produced: %d\n", item);
        in = (in + 1) % BUFFER_SIZE;
        sem_post(&mutex);
        sem_post(&full);
        sleep(1);
    }
}
-- INSERT --
```

rashiga@localhost:~ — vi prodcons.c

```
void* consumer(void* arg) {
    int item;
    for(int i = 0; i < 10; i++) {
        sem_wait(&full);
        sem_wait(&mutex);
        item = buffer[out];
        printf("Consumed: %d\n", item);
        out = (out + 1) % BUFFER_SIZE;
        sem_post(&mutex);
        sem_post(&empty);
        sleep(2);
    }
}
int main() {
    pthread_t p, c;
    sem_init(&mutex, 0, 1);
    sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);
    pthread_create(&p, NULL, producer, NULL);
    pthread_create(&c, NULL, consumer, NULL);
    pthread_join(p, NULL);
    pthread_join(c, NULL);
-- INSERT --
```